Q1: Time Taken: 4 minutes

Ans: Quantum computing can break traditional publickey cryptosystems like RSA and ECC using Shor's algorithm, which can quickly factor large numbers and solve discrete logarithm problems. This makes current encryption methods insecure in the future.

To replace them, we use Post-Quantum Cryptography (PQC) algorithms such as—

* Lattice-based cryptography (e.g., CRYSTALS-Kyber, Dilithium)
* Code based cryptography (e.g., McEliece)
* Hash-based signatures (e.g., SPHINCS+)

These algorithms resist quantum attacks because they are based on mathematical problems like lattice problems or error-correcting codes, which are not efficiently solved by known quantum algorithms like Shor's algorithm.

Thus, post-quantum algorithms provide security even against quantum computers.

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Q2: Time Taken: 5 min and 30 sec.

Ans: ~~ID~~

Idea:

The generator uses—

* current timestamp (time.time())
* Process ID (os.getpid())
* Modulus operation to limit the range

These values are combined and updated each time to generate pseudo-random numbers.

Python Implementation:

Python

```
Import time
Import os
class SimplePRNG:
    def_ init_ (self, mod = 100):
        # initial seed using
timestamp and process ID
        self. mod = mod
        self. seed =
int(time. time()) + os. getpid()
```

```
def next (self):
    # Linear formula to generate next number
  self.seed = (self.seed * 1103515245 + 12345) % (2**31)
    return self.seed % self.mod
  # Limit output range

  # Example usage
  prng = SimplePRNG(mod = 50)
  # Number between 0-49

  for _ in rang (5):
  print (prng.next ())
```

Explanation:

The timestamp and process ID create an initial unpredictable seed.

A linear congruential formula updates the seed.

The modulus operation ensures the output stays within the desired range.

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Q3: Time Taken: 5 minutes

Ans:

Traditional ciphers such as Caesar, Vigenere and Playfair are classical encryption methods. They use simple substitution or shifting techniques with very short keys. Their main strength is simplicity and ease of implementation. However, they are weak against modern cryptanalysis techniques like frequency analysis and brute force attacks. They provide very low security and are not suitable for digital communication. Encryption and decryption are fast but insecure.

Modern symmetric ciphers like DES, and AES use complex mathematical operations and much longer keys (DES: 56-bit, AES: 128/192/256-bit). They are designed for computers and provide high security against brute-force and statistical attacks. AES is especially strong and widely today. These ciphers are also efficient and fast in hardware and software implementations.

In summary, traditional ciphers are simple but insecure, while modern symmetric ciphers offer strong security, longer keys and resistance to modern cryptanalysis.

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Q4: Time Taken: 5 minutes

Ans:

Let S_4 be the symmetric group on 1, 2, 3, 4.

Let X be the set of all 2-element subsets of 1, 2, 3, 4.

1. Define the Action:

for $\sigma \in S_4$ and a subset $A = a, b \in X$, define:

$\sigma \cdot \{a, b\} = \{\sigma(a), (b)\}$

2. Well-defined:

Since σ is a permutation, it maps elements of 1, 2, 3, 4 to elements of the same set.

Thus, $\sigma(a)$, $\sigma(b)$ is also a 2-element subset.

so the action is well-defined.

3. Orbit of 1, 2:

All 2-element subsets are reachable by permutations.

Number of 2-element subsets:

$\binom{4}{2} = 6$

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4. Stabilizer of 1, 2:

Stabilizer = permutations that keep 1, 2 unchanged

we can:

* Swap 1 and 2 (2 ways)
* Swap 3 and 4 (2 ways)

Total = $2 \times 2 = 4$.

So, stabilizer size = 4.

(Also satisfies Orbit-Stabilizer theorem:

$|S_4| = 24 = 6 \times 4$).

Q5: Time Taken: 4 minutes

Ans:

$GF(2^2)$ is constructed using the irreducible polynomial

$x^2 + x + 1$ over $GF(2)$.

The elements of $GF(2^2)$ are:

$\{0, 1, \alpha, \alpha + 1\}$

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(i) Group under multiplication:

The nonzero elements are:

$\{1, \alpha, \alpha+1\}$

* multiplication is closed.
* Identity element is 1.
* Every nonzero element has an inverse.

Hence, the nonzero elements of $GF(2^2)$ form a group under multiplication.

(ii) Cyclic property

We check powers of α:

$a^1 = \alpha$

\alpha^2 = \alpha + 1

$a^3 = 1$

Since powers of α generate all nonzero elements, the group is cyclic.

Therefore, the nonzero elements of $GF(2^2)$ form a cyclic group under multiplication.

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Q6: Time Taken: 3 minutes

Ans:

Let
$Z = \{ \lambda I \mid \lambda \in R, \lambda \neq 0 \}$

1. Subgroup:

The product of two scalar matrices is scalar:

$(\lambda I)(\mu I) = (\lambda \mu) I$

So, Z is a subgroup.

2. Normality:

For any $A \in GL(2, R)$,

$A(\lambda I)A^{-1} = \lambda$.

3. Factor group:

The factor group is

$GL(2, R)/Z = PGL(2, R)$

Interpretation:

This group represents invertible matrices up to scalar multiples (ignoring scaling).

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Q7: Time Taken: 5 minutes

Ans:

Diffie-Hellman Key Exchange:

The Diffie-Hellman key exchange is a method that allows two users to generate a shared secret key over a public channel.

1. Both users agree on a large prime number p and a generator g.

2. Each user chooses a private key and computes a public value.

3. They exchange public values and compute the same shared secret key.

Security:

The security is based on the discrete logarithm problem, which is hard to solve for large prime numbers. An attacker can not easily find the private key from the public value.

However, it is vulnerable to a man-in-the middle attack if authentication is not used.

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If the prime modulus is small:

If p is not large enough, attackers can solve the discrete logarithm problem easily and break the key.

So, a large prime number is necessary for strong security.

Thus, Diffie-Hellman provides secure key exchange when large primes and authentication are used.

Q8: Time Taken: 4 minutes.

Ans:

Let H_1 and H_2 be two subgroups of a group G.

Proof:

Consider $H = H_1 \cap H_2$

1. Identity:

Since both H_1 and H_2 contain the identity element e, their intersection also contains e.

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2. Closure:

If $a, b \in H$, then $a, b \in H_1$ and $a, b \in H_2$.

Since both are subgroups $ab \in$ and $ab \in H_2$.

so, $ab \in H$.

3. Inverse:

If $a \in H$, then $a \in H_1$ and $a \in H_2$,

So $a^{-1} \in H_1$ and $a^{-1} \in H_2$.

thus, $a^{-1} \in H$.

Therefore, $H_1 \cap H_2$ is a subgroup of G.

Example:

Take the group $(\mathbb{Z}, +)$.

Let

$H_1 = 2\mathbb{Z}$ (even integers)

$H_2 = 3\mathbb{Z}$ (multiples of 3)

Then,

$H_1 \cap H_2 = 6\mathbb{Z}$

which is also a subgroup of $(\mathbb{Z}, +)$.

Hence, the intersection of two subgroups is always a subgroup.

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Q9: Time taken: 4 minutes.

Ans:

Let $\mathbb{Z}_n = 0, 1, 2, \ldots, n-1$ with addition and multiplication modulo n.

1. Commutativity:

* addition: $a+b = b+a \pmod{n}$
* multiplication: $a \cdot b = b \cdot a \pmod{n}$

So, $\mathbb{Z}_n$ is a commutative ring.

2. Zero Divisors:

* If n is composite, there exist nonzero $a, b \in \mathbb{Z}_n$ such that $a \cdot b \equiv 0 \pmod{n}$.
* If n is prime, no such $a, b \neq 0$ exist.

3. Field Condition:

* $\mathbb{Z}_n$ is a field if and only if n is prime.
* In this case, every nonzero element has a multiplicative inverse modulo n.

Summary:

* Always commutative
* Has zero divisors if n is composite
* Is a field if n is prime.

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Q10: Time Taken: 4 minutes.

Ans:

Vulnerabilities of DES:

* DES uses a 56-bit key, which is too short for modern computers.

* This makes it vulnerable to brute-force attacks, where all possible keys are tried to decrypt data.

* Some cryptanalytic attacks (like differential and linear cryptanalysis) also weaken DES.

Impact of Key length:

* Short key length allows attackers to try all possible keys quickly, breaking the encryption.

* Longer keys make brute-force attacks infeasible.

AES Improvements:

* AES supports 128, 192, and 256-bit keys, making it much stronger against brute-force attacks.

* AES is designed to resist known cryptanalytic attacks, unlike DES.

Thus, AES replaced DES to provide stronger security and better protection for modern applications.

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